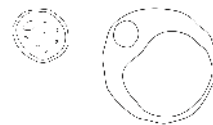




Turmeria Zen™

The emotional wellbeing manager

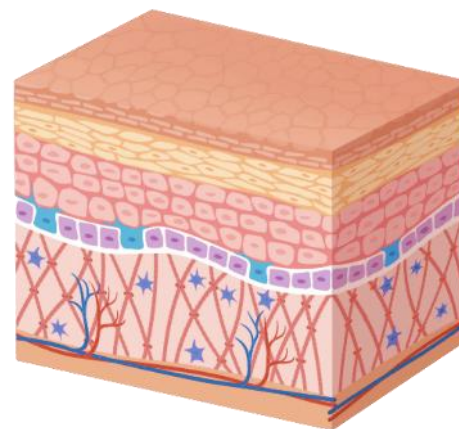
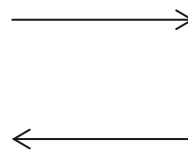




The skin reflects our emotions. Why?

Did you know that in the embryo, the same cells that made your brain also made your skin?

If you feel good, you look good. If you are stressed, your skin will reflect it!

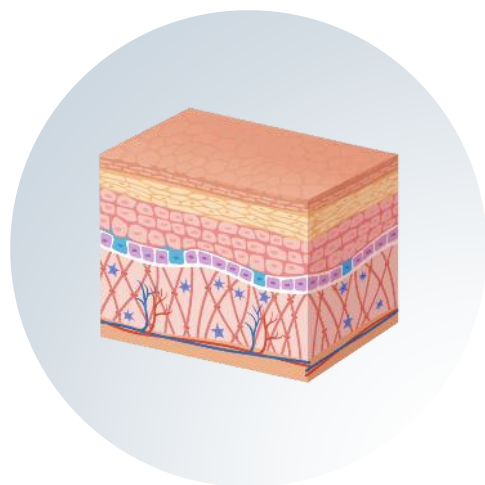




Cosmetic approaches to skin care

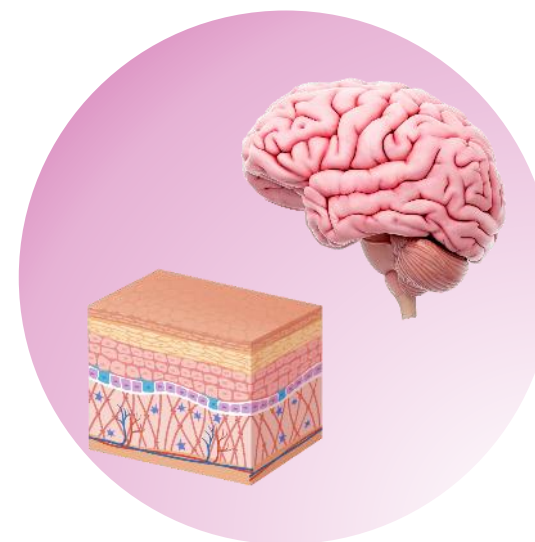
TRADITIONAL COSMETIC APPROACH

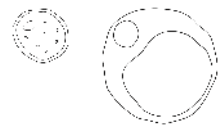
External factors
Physiological conditions
Molecular interactions
Biomedical causes



HOLISTIC COSMETIC APPROACH

Traditional approach, plus:
Emotions
Psycho-social interactions
Psycho-dermatology



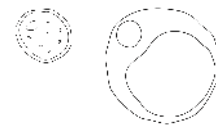


Effects of stress on our skin



Chronic stress





Effects of stress on our skin

Acute and chronic stress causes an emotional state in which many hormones are not well regulated.

Cortisol is one of the key hormones in the stress response, and when accumulated on skin, it can cause physiological disruptions.

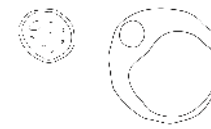
CORTISOL

Hormone which plays a key role as a stress mediator

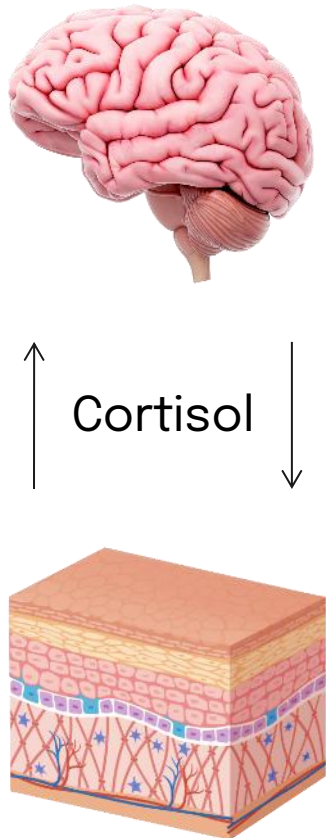


Acute stress





Excess of cortisol affects the skin homeostasis



Reduction on the skin barrier function & water retention capacity (dehydration)

Excessive skin inflammation (sensitization)

Decrease in cell regeneration (wound healing capacity*)

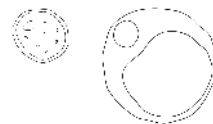
Collagen and elastin (ECM) degradation

Skin microbiota dysbiosis



(*) Walburn J. et al, 2009, "Psychological stress and wound healing in humans: A systematic review and meta-analysis", *J. Psychosomatic Res.*, 67 (3), 253-271





Skin hydration: a crucial pivotal role

One of the most affected skin properties in the stressed brain-skin connection are the epidermal barrier and the skin hydration.

CORTISOL AFFECTS THE SKIN BARRIER

Decreased proliferation of keratinocytes and fibroblasts

Impaired epidermal differentiation

Reduced synthesis of epidermal lipids and dermal hyaluronic acid (HA)

DECREASE OF SKIN HYDRATION

Keratinization disorders:

- Less desquamation
- Hyper-keratinization
- Loss of 3D keratin structure

PREMATURE SIGNS OF AGEING

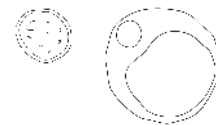
Skin atrophy

Reduction of epidermal thickness

Flattening of the dermo-epidermal junction (DEJ)

Disruption of the dermal fibres network (ECM)





Effects of stress on our skin

PREMATURE SIGNS OF AGEING

Skin dryness	Wrinkles
Skin fading	Dark circles
Elasticity loss	Puffy eyebags

NEUROINFLAMMATORY CONDITIONS

Erythema	Pruritus
Acne	Psoriasis
Vitiligo	Atopic Dermatitis
Itching	



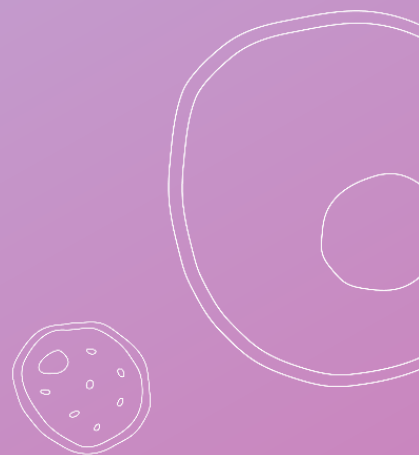


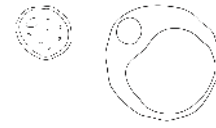
Skin speaks for you and reflects your emotions



The skin, like the brain, is a complex organ affected by emotions and psycho-social interactions such as stress.

If stress is the cause,
counteract it with relax,
positive emotions and
TURMERIA ZEN™.





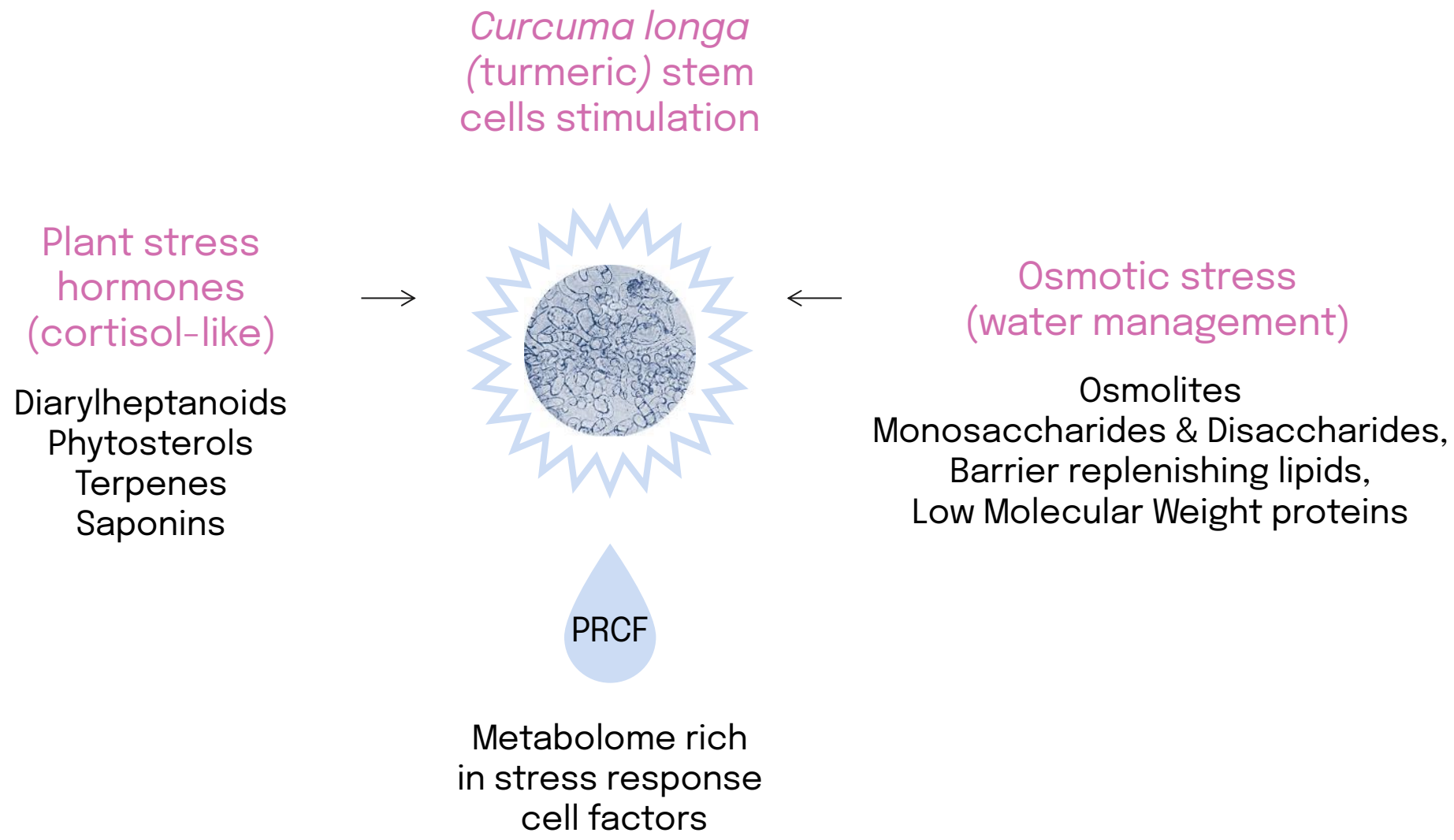
The plant: Turmeric (*Curcuma longa*)

- *Curcuma* has been used for centuries in Ayurvedic and traditional Chinese medicines
- Turmeric is the most studied plant biomedically
- Multiple properties (antioxidant, anti-inflammatory, wound healing, antimicrobial, etc.)





The plant: Turmeric (*Curcuma longa*)





TURMERIA ZEN™

Metabolome rich in water management compounds

ENHANCE THE WATER RETENTION CAPACITY

Total polysaccharides 10000 ppm,
including monosaccharides (glucose, fructose
& *myo*-inositol) and
disaccharides (sucrose)

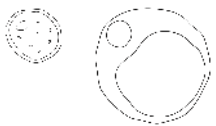
Total proteins 100 ppm, including Low
Molecular Weight proteins

IMPROVE THE SKIN BARRIER FUNCTION

Increase the lipid synthesis and the
keratinocytes proliferation

Free Fatty Acids (FFAs),
including palmitic acid,
alpha-linolenic acid (ω -3),
and linoleic acid (ω -6)

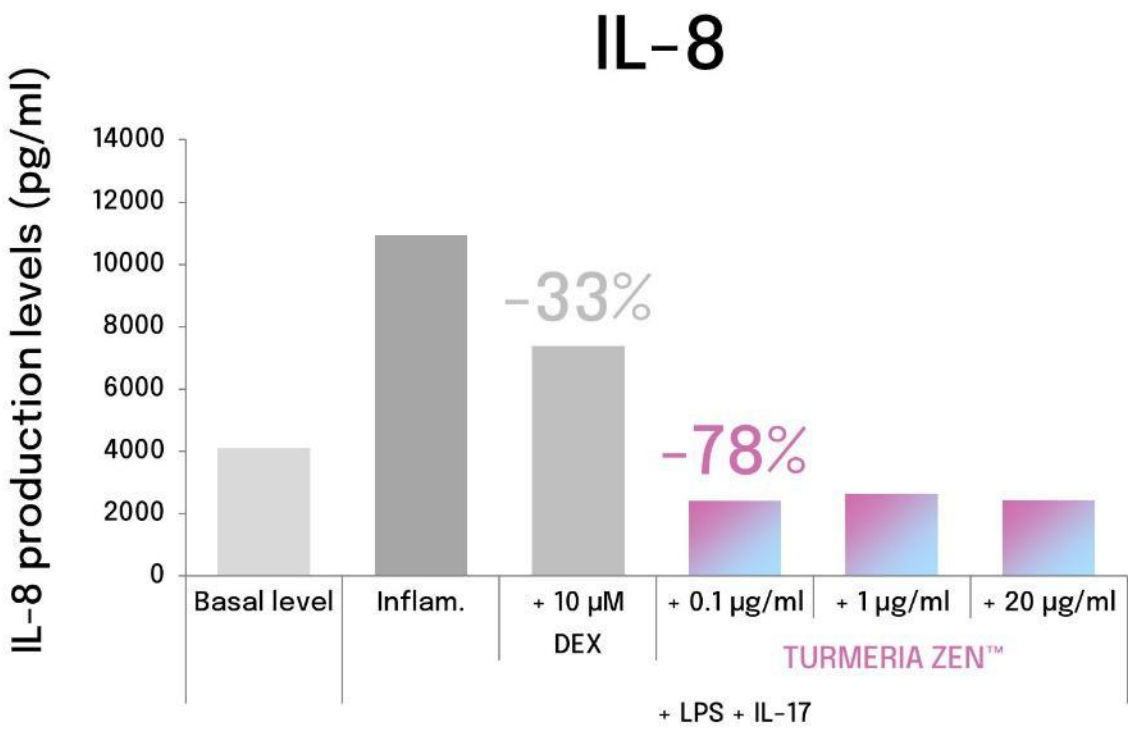
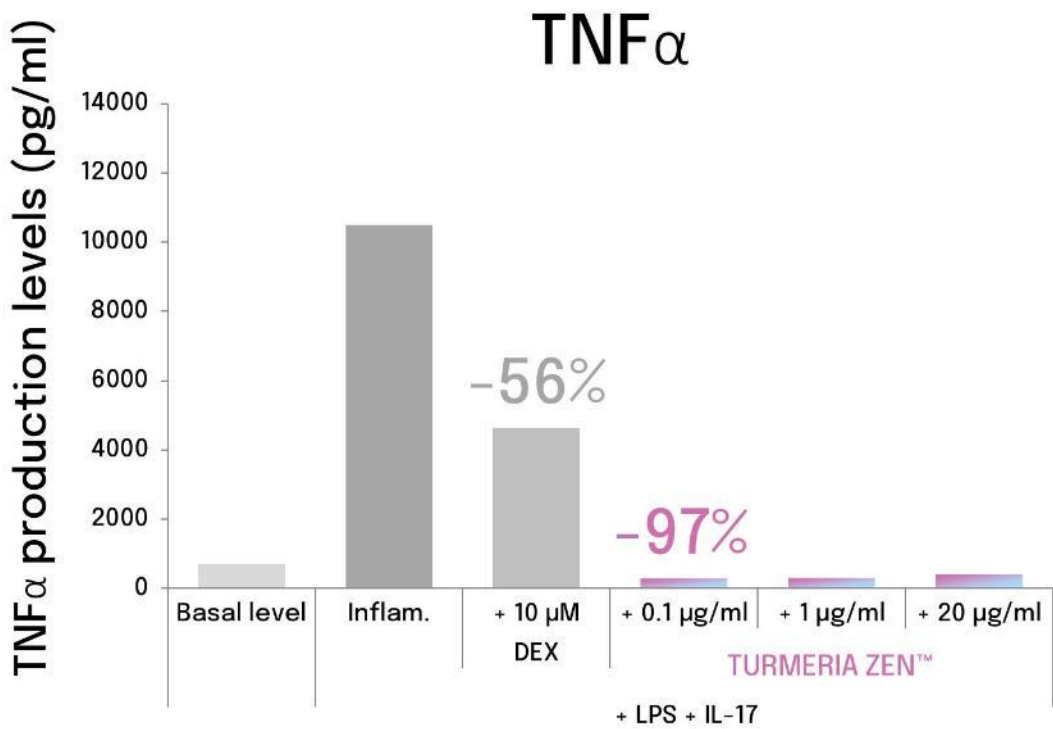




In vitro efficacy

1. Anti-inflammatory activity

MINDFULNESS* EFFECT



- Human Monocytes THP-1.
- Cells were incubated with 10 μ g/ml LPS (Lipopolysaccharide) & 3 ng/ml IL-17 to induce inflammation, and then they were incubated for 24 h in the absence or presence of the test product. Monocytes not treated with LPS were used as the negative control. 10 μ M Dexamethasone (DEX) was used as a positive control.
- TNF- α and IL-8 levels were quantified by ELISA.

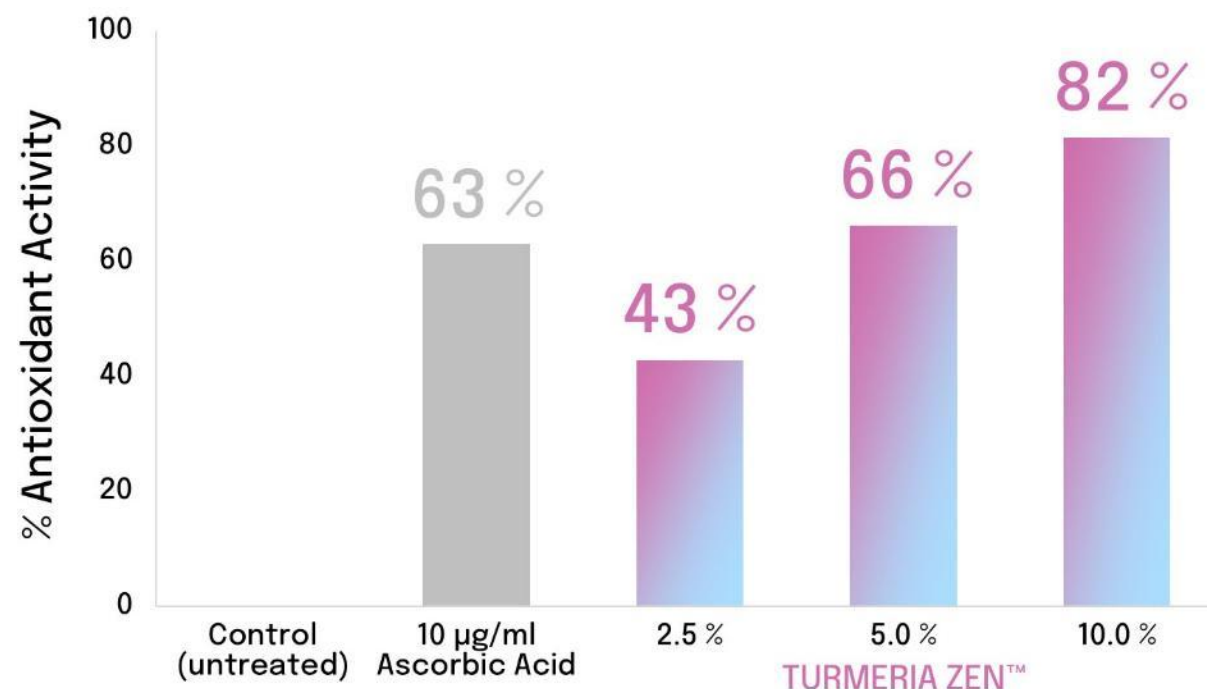
*Brain Behav. Immun. 2013; 27C: 174-184





In vitro efficacy

2. Antioxidant activity | DPPH Assay



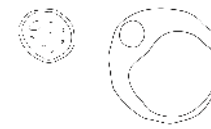
UP TO 82%

TURMERIA ZEN™ was able to capture up to an 82% of the free radical DPPH, showing an antioxidant effect



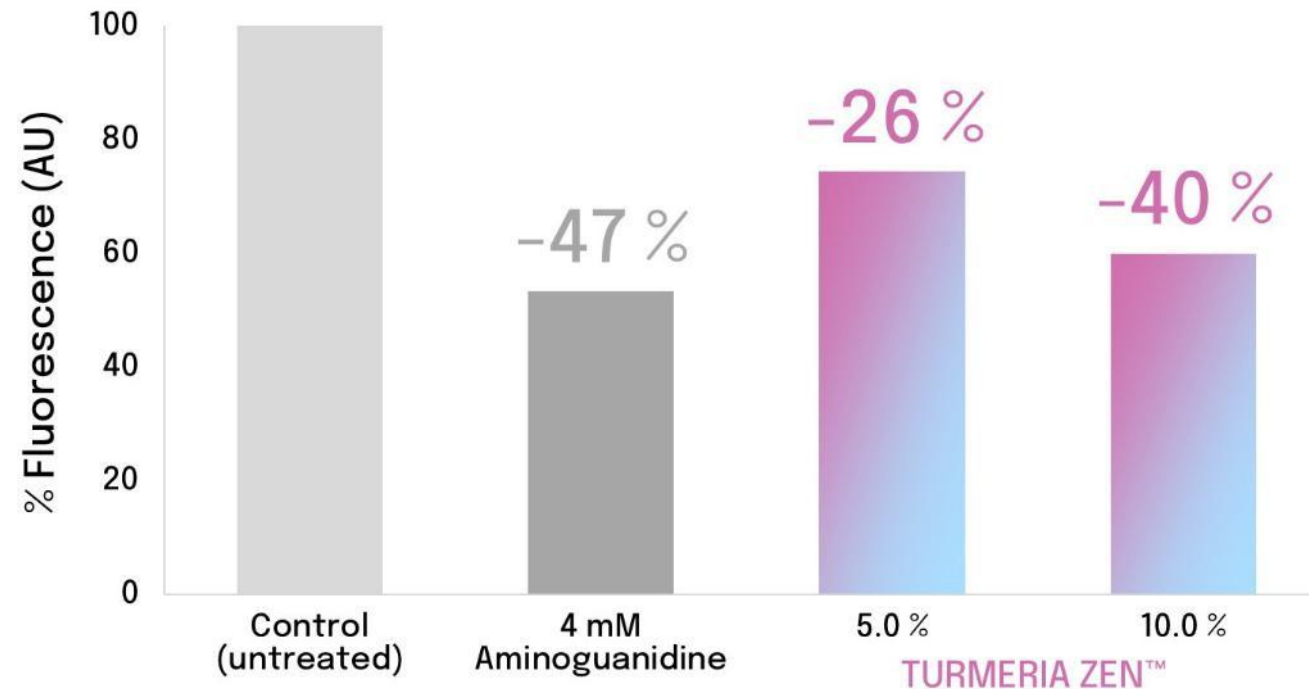
- 10 min reaction with DPPH
- Measurements of A(517 nm)
- Higher A(517 nm) means higher antioxidant effect (capacity to capture the DPPH)





In vitro efficacy

3. Anti-glycation activity



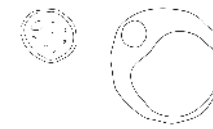
DOWN TO 40%

TURMERIA ZEN™ was able to inhibit the glycation reaction by up to a 40%, a comparable effect to the positive control



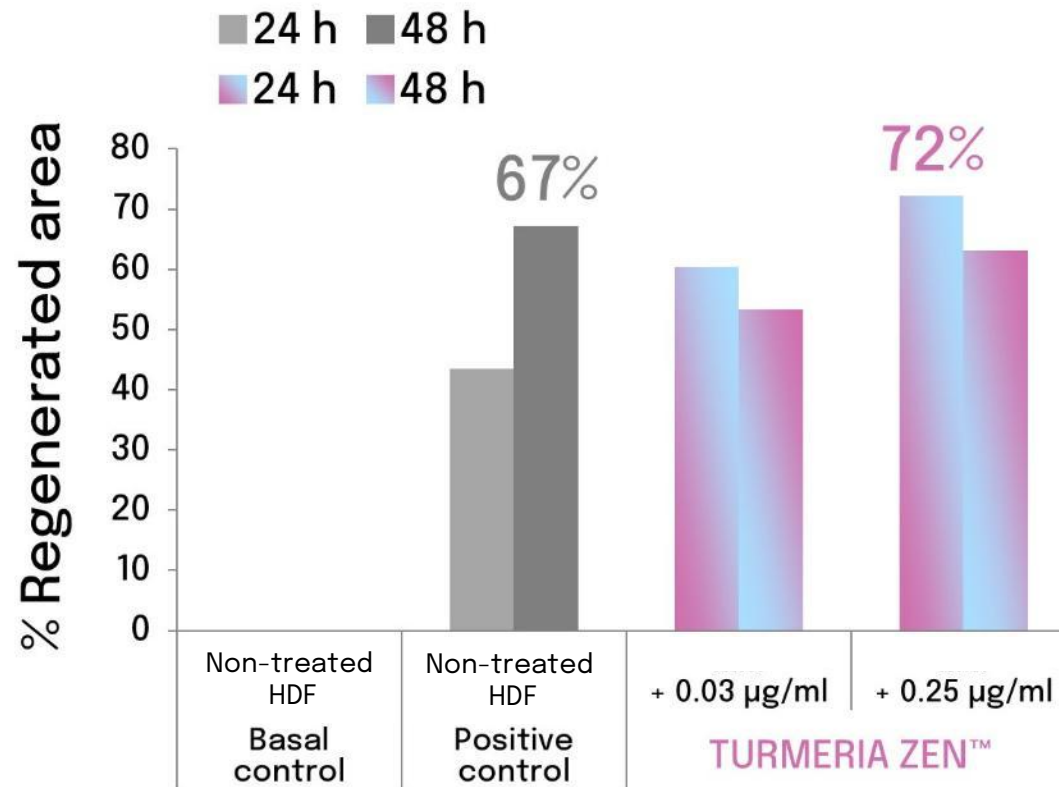
- 2 mg/mL BSA + 40 mM glucose (100% glycation), vs reaction with Aminoguanidine or Turmeria Zen™
- 15 min reaction
- Measurements of fluorescence at 440 nm
- Higher fluorescence means higher glycation reaction





In vitro efficacy

4. Regenerative properties (Scratch test on HDF)

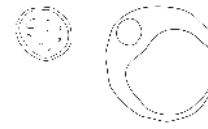


TURMERIA ZEN™ regenerated the area by +72% at 24h vs baseline conditions

The effect was higher than that of the positive control

- Non-treated and treated HDF were cultured in maintenance medium.
- Non-treated HDF were propagated and grown in growth medium.
- The growth medium composition contained benchmark growth factors at different concentrations: basal control at 0.1% and positive control at 10%.

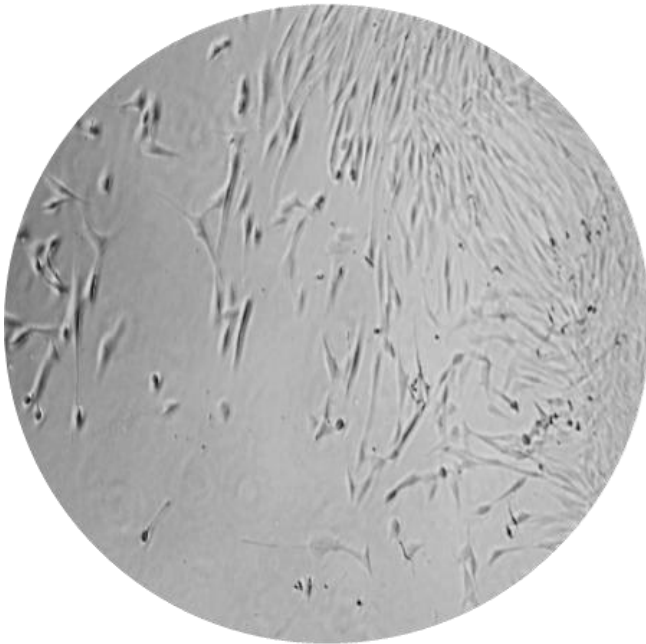




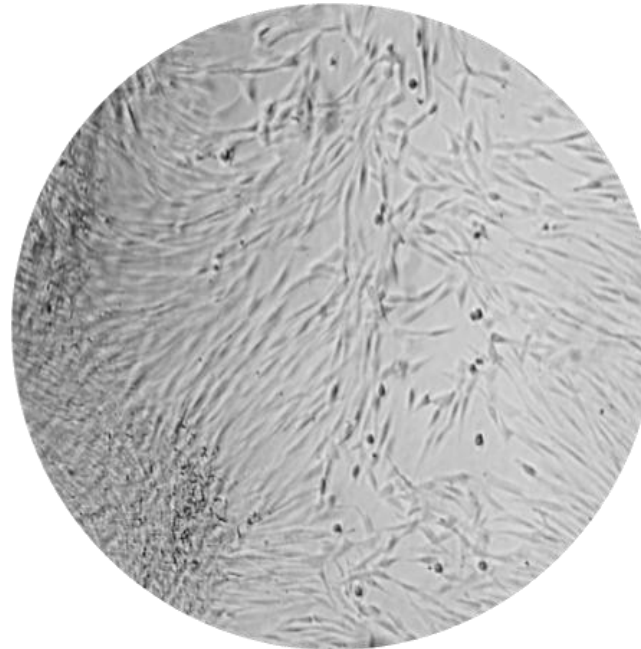
In vitro efficacy

4. Regenerative properties (Scratch test on HDF)

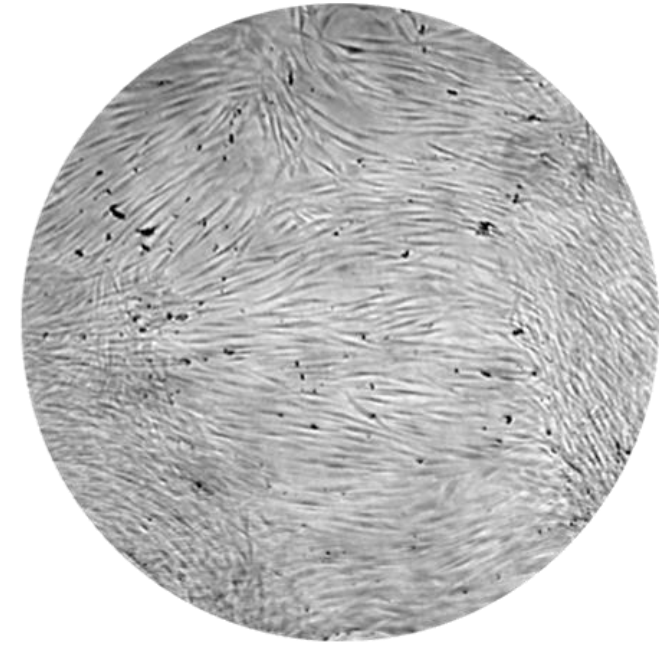
 48h



Basal medium



Positive control



0.25 µg/ml
TURMERIA ZEN™

- Basal medium and Positive control were propagated and grown in growth medium.
- The growth medium composition contained benchmark growth factors at different concentrations: basal medium at 0.1% and positive control at 10%.





Ex vivo efficacy

Work performed with Skin Explants treated to simulate stressed skin and its multiple affections:

Ex vivo 1. Biochemically induced stress

Test on explants stressed with high cortisol levels

Ex vivo 2. Significant physical stress

Test challenging the regeneration on skin explants

Ex vivo 3. Extreme physiological stress

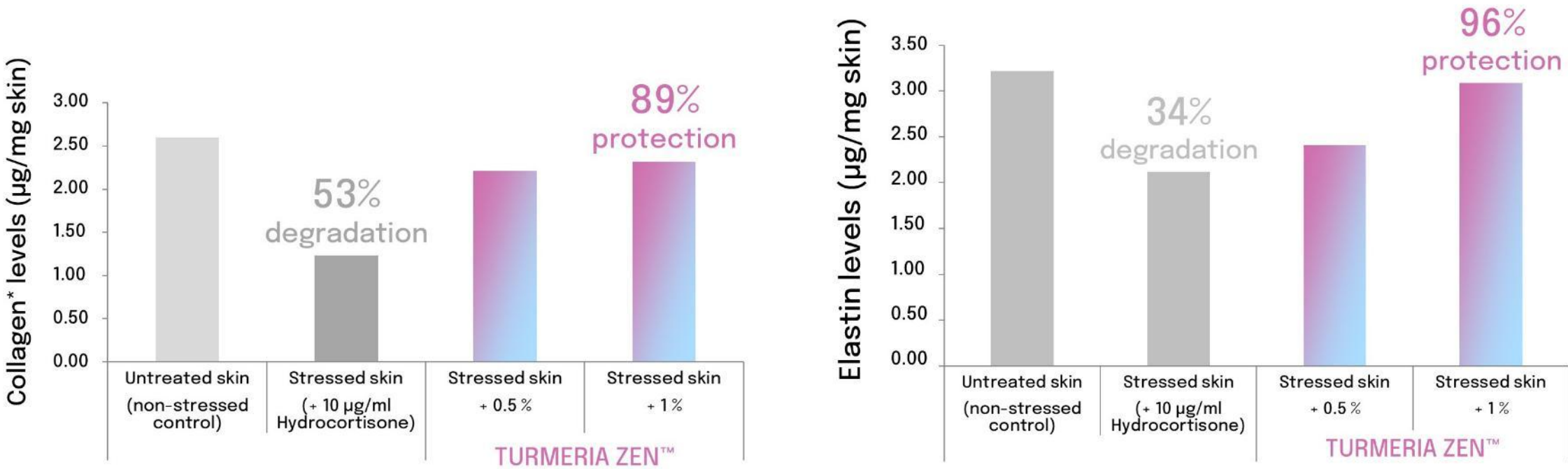
Test on psoriatic and atopic dermatitis skin explants





Ex vivo efficacy

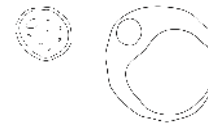
1. Protection of cortisol-stressed explants



(*mainly Collagen Type I)

- Human Organotypical Skin Explant Cultures (hOSECs)
- 12 days culture
- 10 µg/ml Hydrocortisone was used to induce stress
- Untreated hOSECs were used as the negative control
- 0.5% and 1% Turmeria Zen™ were added together with Hydrocortisone to study the active's protective effect

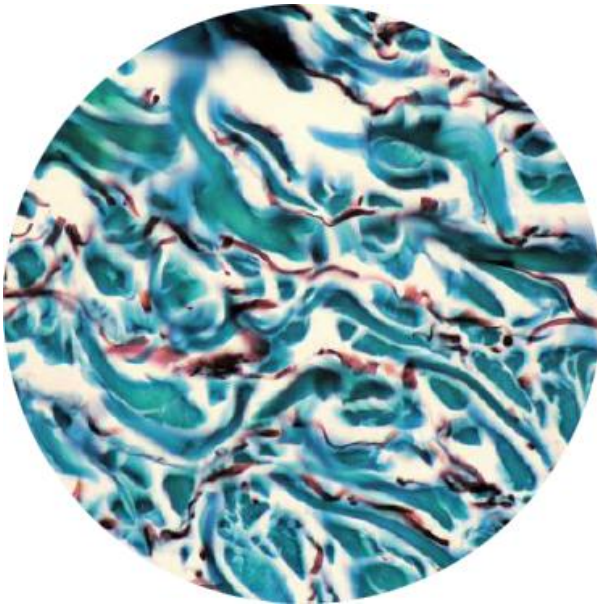




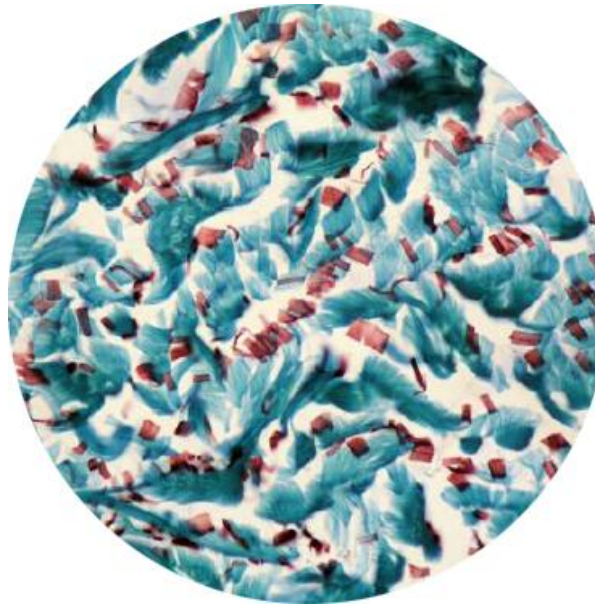
Ex vivo efficacy

1. Protection of cortisol-stressed explants

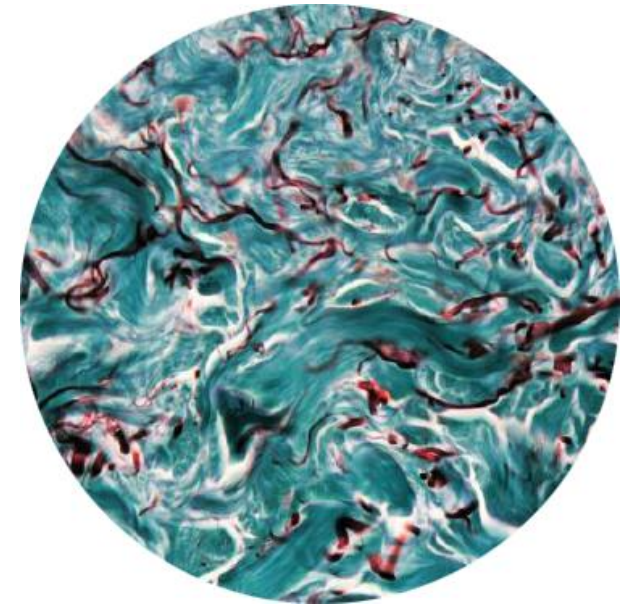
Non stressed skin



Stressed skin
(10 µg/ml
hydrocortisone daily)



Stressed skin +
1% TURMERIA ZEN™



- Blue staining: Collagen fibres ; Red staining: Elastin fibres
- White gaps: interstitial space between fibres
- Daily stress breaks the collagen and elastin fibres into smaller fragments, but daily stress + Turmeria Zen™ maintains and redensifies the ECM (extracellular matrix)



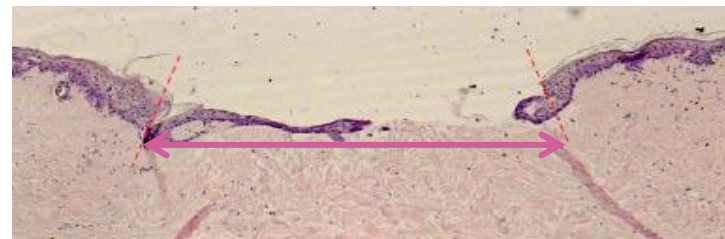
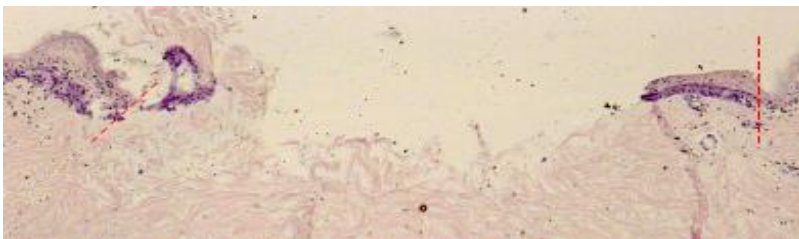
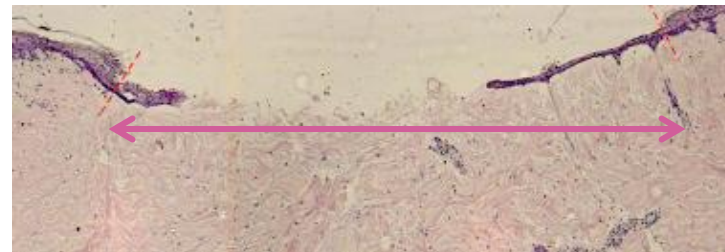
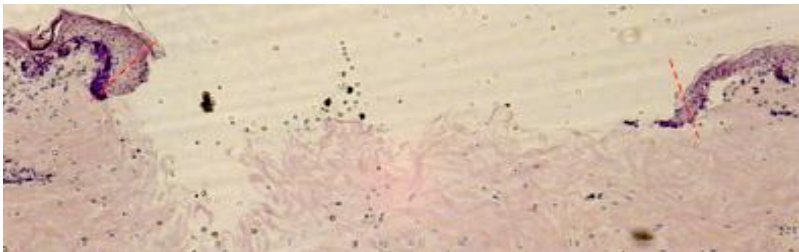
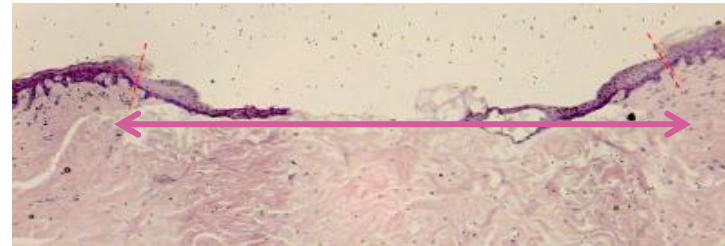
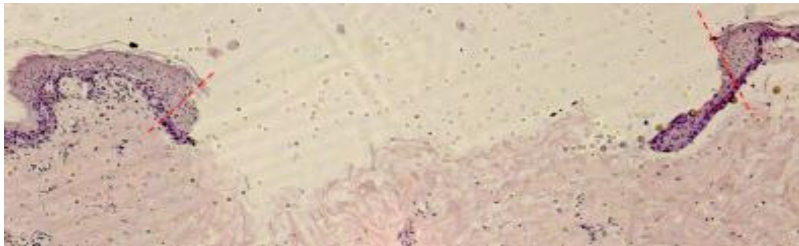


Ex vivo efficacy

2. Regenerative properties (Punch biopsies)

5 µg/ml EGF
2-fold vs untreated

50 µg/ml TURMERIA ZEN™
2.6-fold vs untreated

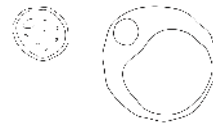


TURMERIA ZEN™ was able to regenerate the epidermis, re-densifying and restructuring the skin more than EGF



- Skin biopsies from human donors. 8 mm discs with 3mm inner discs where the epidermis was removed (punch)
- 8 days culture with 5 µg/ml EGF or 50 µg/ml Turmeria Zen™





Ex vivo efficacy

3. Skin biopsies from psoriasis & atopic dermatitis (AD)

- Full-thickness biopsies from atopic dermatitis & psoriasis as models of inflamed and stressed epidermis
- Study carried out in the unit of Translational Immunology (PCB/UB)
- Approved by the ethical committee of the clinical center





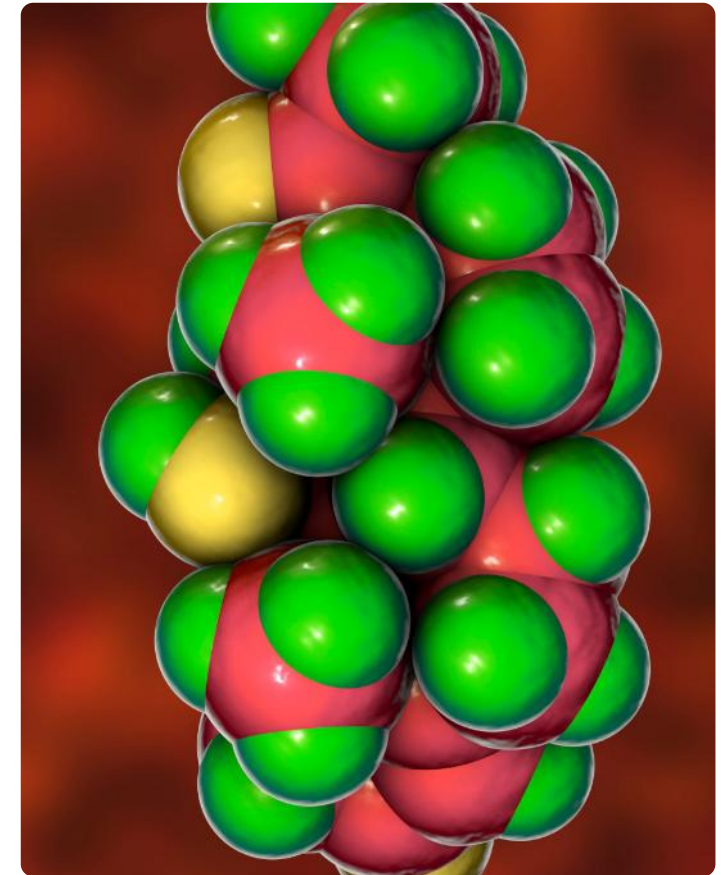
Ex vivo efficacy

3. Skin biopsies from psoriasis & atopic dermatitis (AD)

Correlation between stress, high levels of salivary cortisol and psoriasis/AD

DID YOU KNOW THAT...?

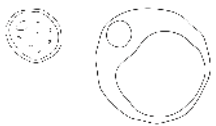
- Salivary cortisol levels are higher in patients with psoriasis compared to healthy subjects¹
- Salivary cortisol levels are significantly higher in AD patients compared to healthy subjects²



1. Rousset L. et al., 2018, 'Stress and psoriasis', *Int J Dermatol*, 57, 1165-1172

2. Megumi M. et al., 2013, 'Stress Evaluation in Adult Patients with Atopic Dermatitis Using Salivary Cortisol', *BioMed Res Int*, Vol. 2013



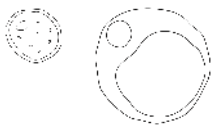


Ex vivo efficacy

3.1 Skin biopsies from psoriatic skin

mRNA	ROLE IN PSORIASIS
IL-23A	It is associated with psoriasis severity
CXCL10	IFN-μ-dependent cytokine. It is significantly higher in patients with psoriasis
IFN-γ	Psoriasis severity and prognosis marker
DEF4	Beta-defensin 2 is a clinical efficacy marker in biological therapy in psoriasis. Defensins are anti-microbial peptides (AMPs), and therefore they are part of the immune response and contribute to re-equilibrate the unbalanced microbiota, known as dysbiosis
K-16	Keratin 16 is a marker in epidermal hyperplasia and abnormal differentiation in psoriasis. Its diminution is associated to the clinical improvement in all effective treatments in psoriasis (in the psoriatic pathogenesis, there is a hyper-keratinization and a disorganization of the Keratin)



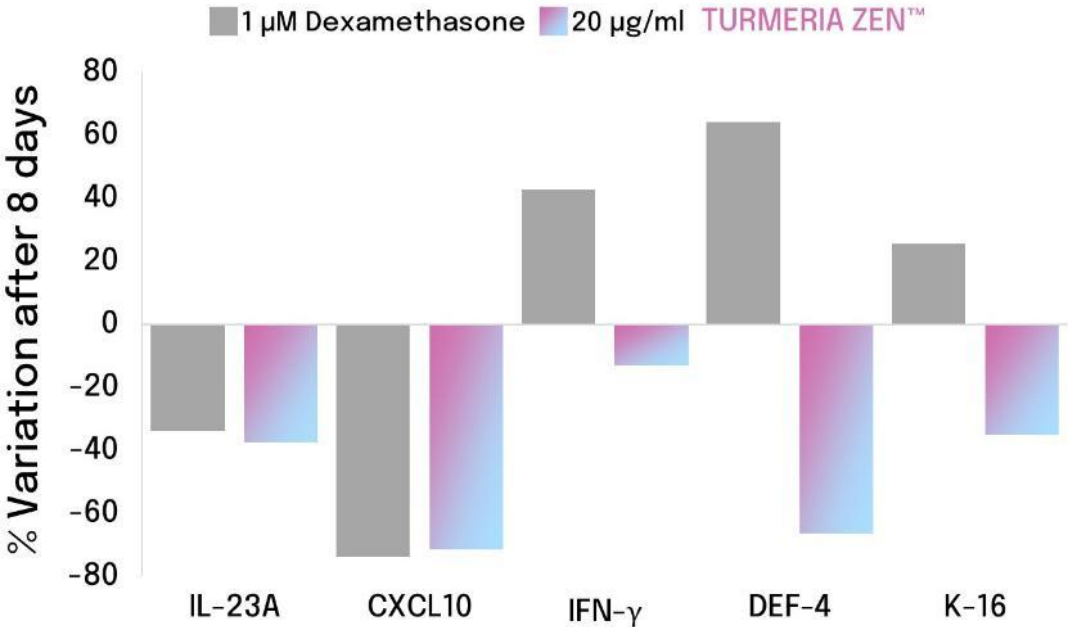


Ex vivo efficacy

3.1 Skin biopsies from psoriatic skin

PATENT
PENDING

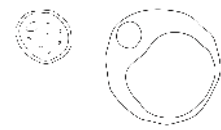
mRNA	Dexam.	TURMERIA ZEN™
IL-23A	- 34 %	- 37 %
CXCL10	- 74 %	- 72 %
IFN-γ	+ 43 %	- 13 %
DEF-4	+ 64 %	- 66 %
K-16	+ 25 %	- 35 %



TURMERIA ZEN™ reduces pro-inflammatory markers, Defensin-4 (anti-microbial peptides (AMPs), and Keratin-16 (avoids hyper-keratinization)

○ Patient 2 (best case of psoriatic skin explant)
○ mRNA levels measured by Quantitative PCR using TaqMan PreAmp Master Mix Kit (Applied Bioscience)



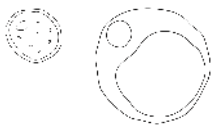


Ex vivo efficacy

3.2 Skin biopsies from atopic dermatitis

mRNA	ROLE IN ATOPIC DERMATITIS
CCL13	Pro-inflammatory chemokine, target of AD treatments
CCL17	Pro-inflammatory chemokine, target of AD treatments
CCL18	Pro-inflammatory chemokine, target of AD treatments
IL-13	Pro-inflammatory cytokine involved in the AD pathogenesis
K-16	Keratin 16 is a marker in epidermal hyperplasia and abnormal differentiation in the AD. Its diminution is associated to the clinical improvement in all effective treatments in AD (in the AD pathogenesis, there is a hyper-keratinization and a disorganization of the Keratin)



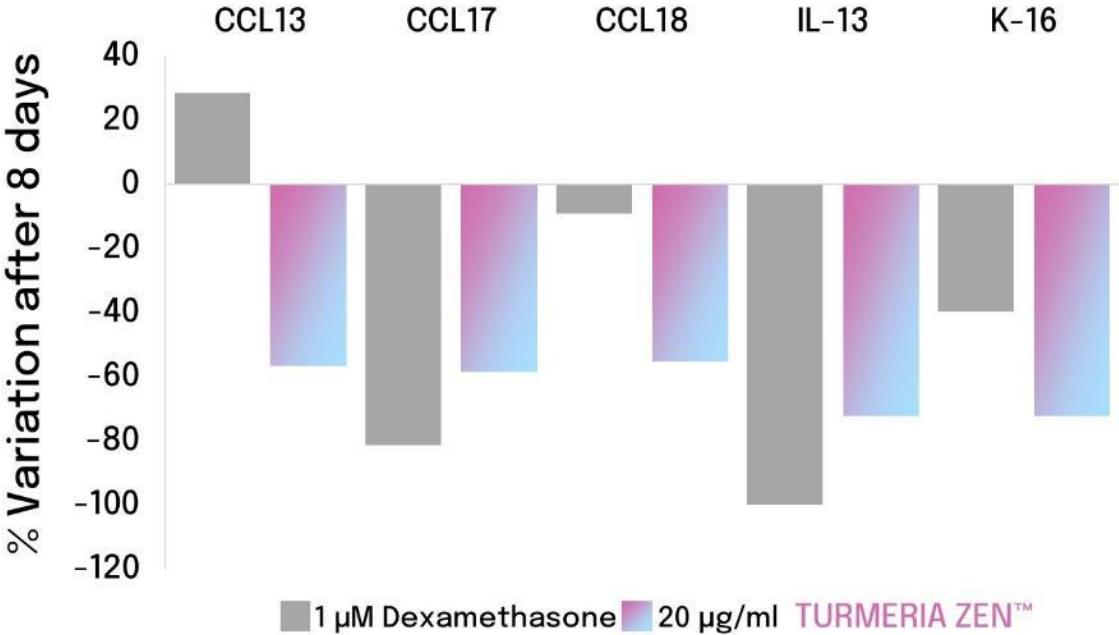


Ex vivo efficacy

3.2 Skin biopsies from atopic dermatitis

PATENT
PENDING

mRNA	Dexam.	TURMERIA ZEN™
CCL13	+ 28 %	- 57 %
CCL17	- 81 %	- 59 %
CCL18	- 9 %	- 55 %
IL-13	- 100 %	- 72 %
K-16	- 40 %	- 72 %



TURMERIA ZEN™ was able to reduce the pro-inflammatory markers CCL13, CCL17, CCL18 &, IL-13, and Keratin-16, preventing the hyper-keratinization

○ Preliminary results – Patient 7 (best case of AD skin explant)
○ mRNA levels measured by Quantitative PCR using TaqMan PreAmp Master Mix Kit (Applied Bioscience)



In vivo test: Well-ageing in stressed profiles

- Pre-selection of 40 vol. STAI questionnaire
- 20 vol. (from those 40) – highest cortisol levels in saliva 30 min. after waking up in the morning (8-35 nM, average 15 nM*)
- Single blind placebo-controlled study. Area of application: face (crow's feet)
- 2 daily applications (28 days, winter –January-February-, in Poland)



1 %
Dosage

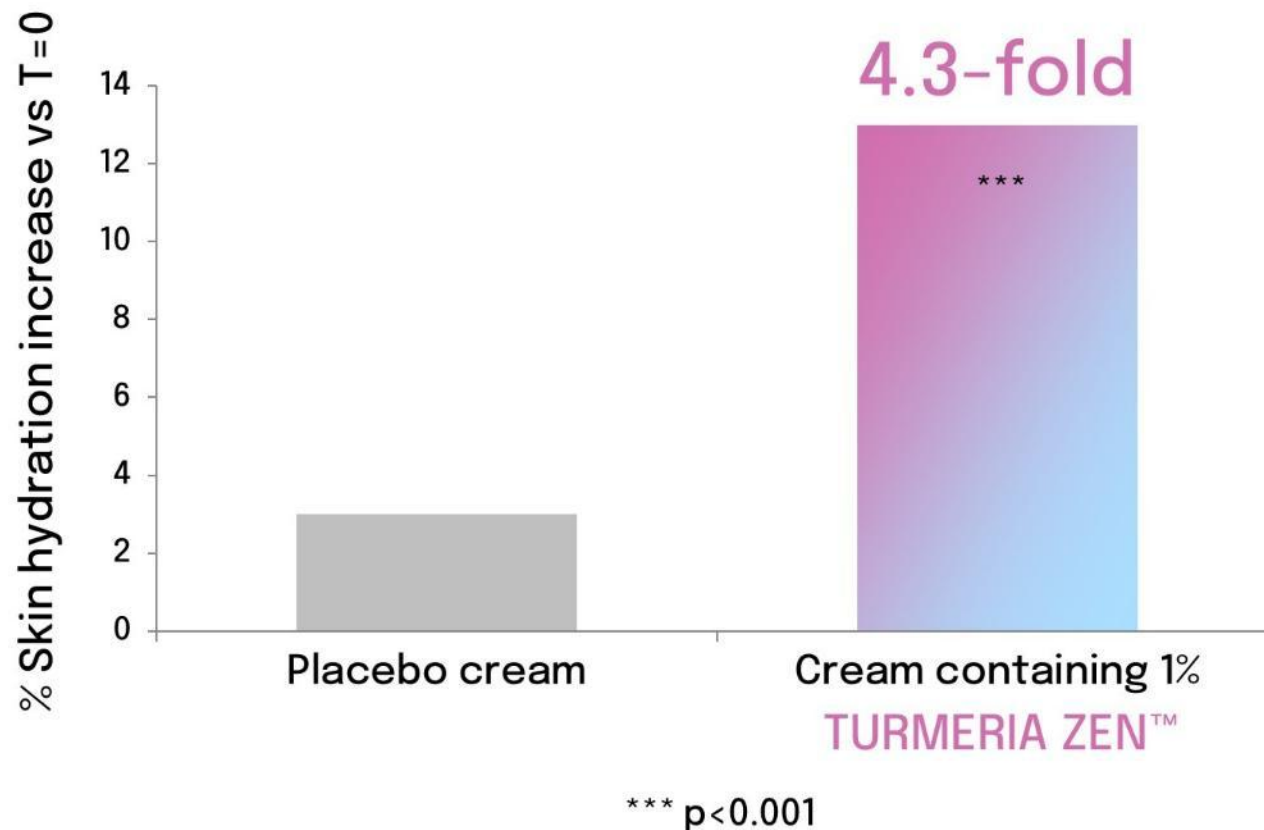
28 Days

35-50 
Years

20 
Volunteers



1. Skin hydration effect



HYDRATION UNDER EXTRA COLD STRESS

TURMERIA ZEN™ significantly increased the skin hydration, by 13% versus initial measurement, i.e., 4.3-fold the placebo effect

- 20 volunteers with high cortisol levels (8-35 nM, average 15 nM). Age: 35-50
- Cream with 1% Turmeria Zen™
- Twice a day face (crow's feet) application for 28 days
- CM 825 Corneometer® (COURAGE & KHAZAKA)





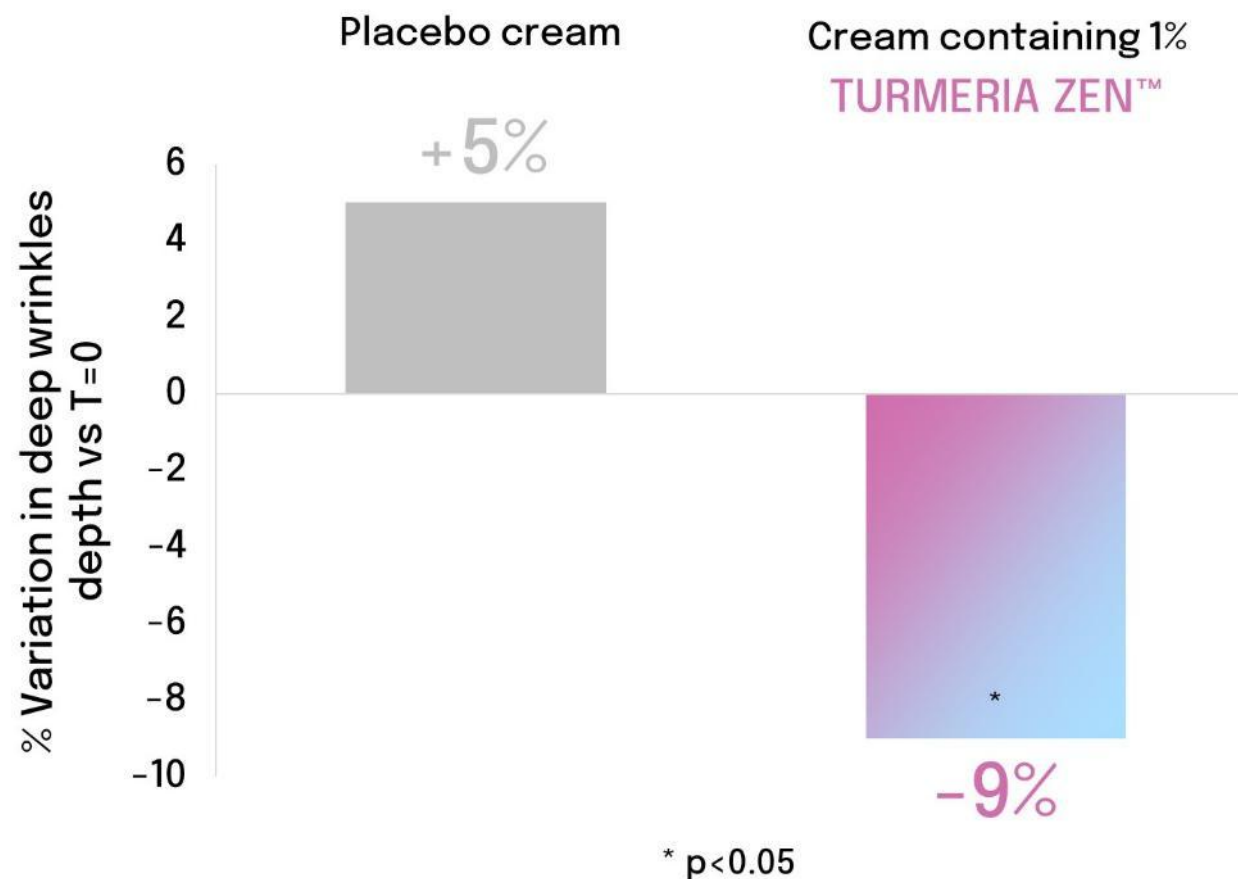
Emotional hydration is
the key for well-being

Longer term action
Holistic approach which
considers skin biology &
brain-skin connection





2. Anti-wrinkles effect

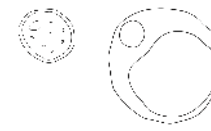


THE ANTI-STRESS WRINKLES,
DOWN TO -40%

TURMERIA ZEN™ significantly reduced the depth of deep wrinkles, by -14% versus placebo at 28 days and by 16 µm on average

- 20 volunteers with high cortisol levels (8-35 nM, average 15 nM). Age: 35-50
- Cream with 1% Turmeria Zen™
- Twice a day face (crow's feet) application for 28 days
- Silicone imprints Silflo®
- SIA® Quantirides®: depth of wrinkles (µm). Partial panel of 16 vol.



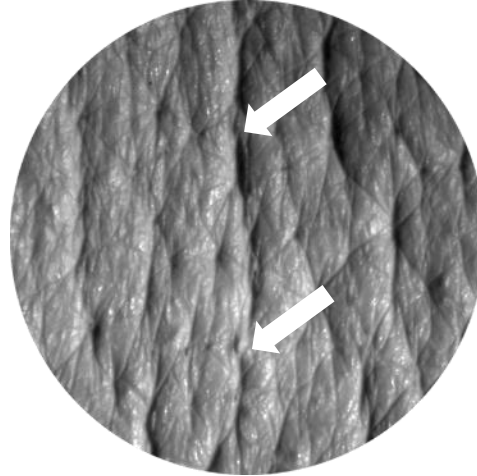
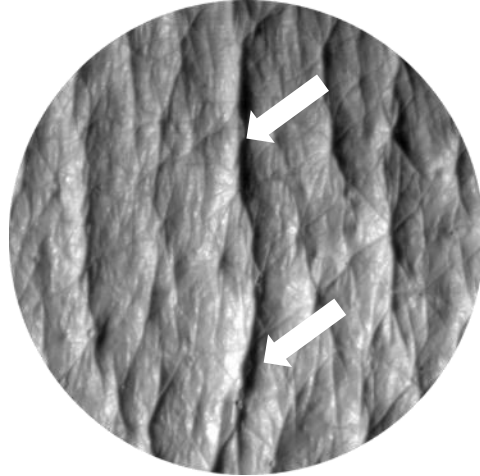


2. Anti-wrinkles effect

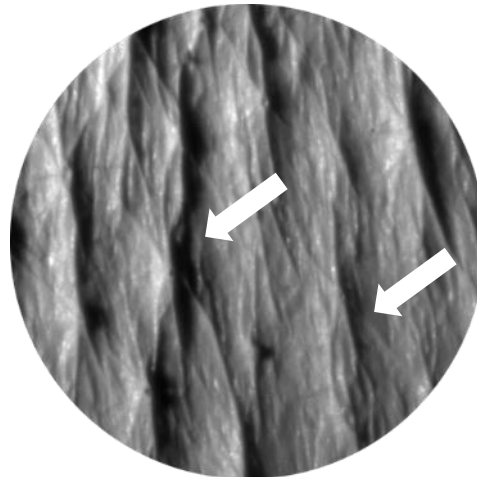
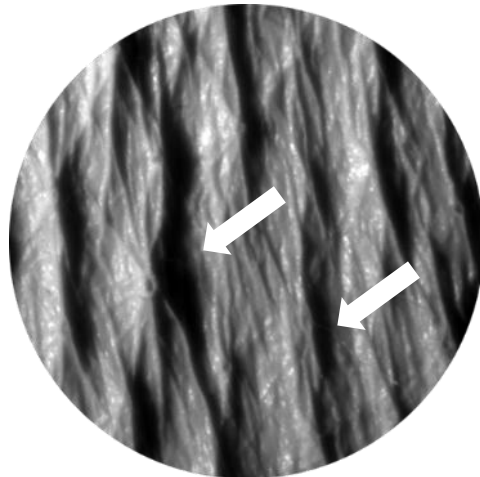
Day 0

After 28 days

Vol.3

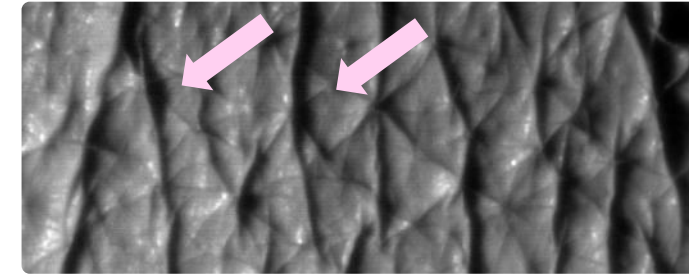


Vol.8

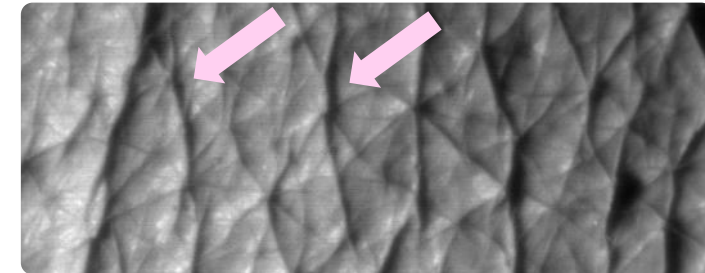


Vol.15

T0d



T28d



Depth reduction of the
deepest wrinkles
($> 110 \mu\text{m}$)



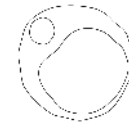
EMOTIONAL COSMETICS

PROTECTING & REGENERATING

ANTI-STRESS WRINKLES
& EMOTIONAL HYDRATION

SLOW COSMETICS & WELL-AGEING





HYDRATION & REGENERATION

Gels, serums, sticks, emulsions, and other formulations

WELL-AGEING & HEALTHY AGEING

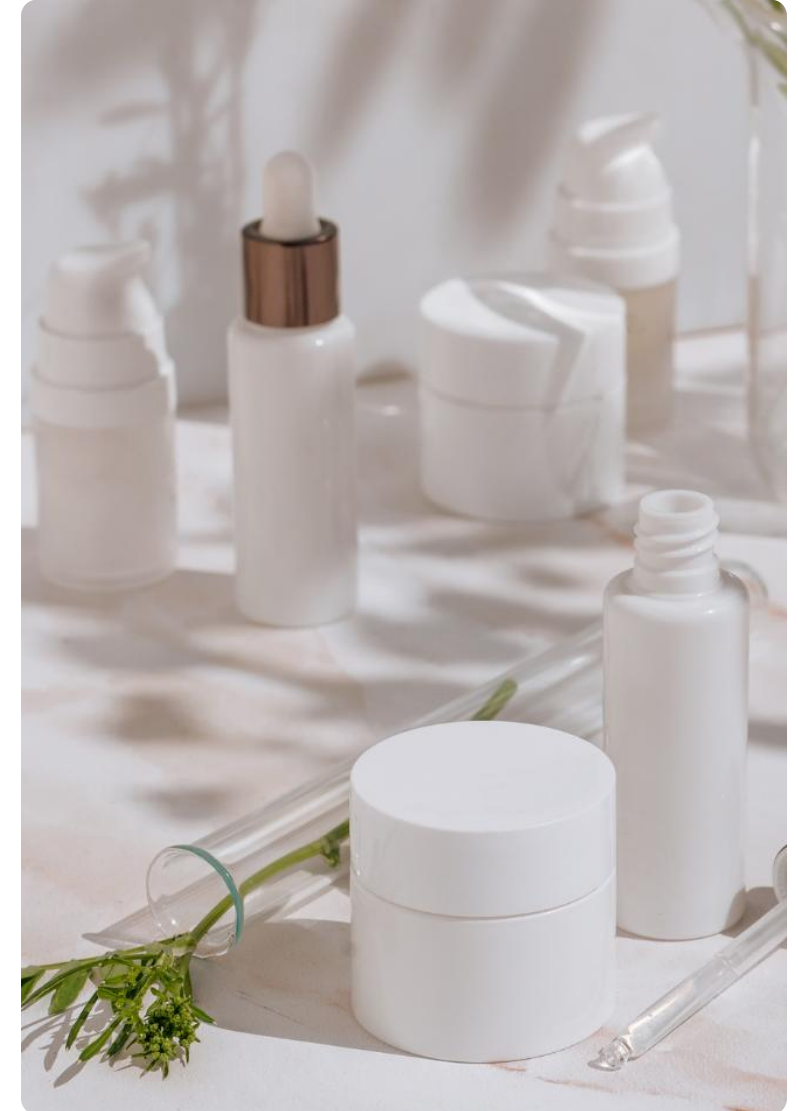
Serums, gels, emulsions and other formulations for the prevention and treatment of wrinkles

SKIN REVITALIZATION AND REGENERATION

Eye contour treatments and mists or serums for scalp regeneration

SLOW BEAUTY & SELF-CARE

Massage lotions; face masks; gels; light creams and serums



Technical information

Declarations

Free from: GMO, BSE, gluten, palm oil, cosmetic allergens*, nanomaterials, VOC

COSMOS approved

ISO 16128 Natural Origin Index: 100%

HALAL Certified: IH-1073/2.1/B/125**

Dosage

Suggested use level: 0.5 – 2%

INCI

Curcuma Longa (Turmeric) Root Extract, Glycerin, Citric acid

Preservative-free

Formulation

Dispersible in water and in ethanol

Incorporation during the cooling phase (<40°C)

*In accordance with Regulation (EC) No. 1223/2009, Annex III, No. 67-92

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Click [here](#) to get access to technical documentation and formulation inspiration



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